# Hands-On-Learning: Baseline or Bust – Your First Reproducible Benchmark

# Learning Objectives

By the end of this activity, learners will be able to:

- Run a lightweight benchmarking harness on a real 15-prompt evaluation dataset with fixed settings.
- Compute and interpret p50/p95 latency, tokens/sec, and cost per task using exported logs.
- Identify outliers and potential bottlenecks in LLM performance from baseline data.
- Formulate data-driven hypotheses for the biggest performance wins to pursue next.



# Dataset Provided

Learners must have a real dataset, so here it is.

**Dataset:** support_chat_eval_v1.jsonl

{"id": 1, "prompt": "My order hasn't arrived yet. Can you check the status?"}
{"id": 2, "prompt": "I forgot my password and can't log in. What should I do?"}
{"id": 3, "prompt": "The app keeps crashing when I try to open it."}
{"id": 4, "prompt": "How do I update my billing information?"}
{"id": 5, "prompt": "Do you offer refunds for accidental purchases?"}
{"id": 6, "prompt": "The live chat agent disconnected. Can you reconnect me?"}
{"id": 7, "prompt": "I was double charged this month. Why?"}
{"id": 8, "prompt": "How do I activate two-factor authentication?"}
{"id": 9, "prompt": "Where can I download my previous invoices?"}

{"id": 10, "prompt": "The confirmation email never arrived."}
{"id": 11, "prompt": "Is there a way to export my account data?"}
{"id": 12, "prompt": "I need help cancelling my subscription."}
{"id": 13, "prompt": "Why is the search function returning no results?"}
{"id": 14, "prompt": "How do I change the language settings?"}
{"id": 15, "prompt": "My payment keeps failing—what can I try?"}

A CSV version will also be included:
support_chat_eval_v1.csv

# Structure of the Activity

# Dataset Information / Resources Required

**Evaluation dataset**

**You will use the provided dataset:**

- datasets/support_chat_eval_v1.jsonl
- (or) datasets/support_chat_eval_v1.csv

# Starter repository files

Provided in the project starter repo:

- run_eval.py
- config.yaml and .env.example
- runs/ folder for outputs

# Tools Used

**Primary language:** Python 3.10+

**Libraries:**

- openai (or a compatible LLM SDK)
- pandas (analysis)

**Other:** Terminal, IDE (VS Code, PyCharm)

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## Setup Requirements

- API key in environment or config
- Python env with required packages installed
- Dataset file exists at:
  - datasets/support_chat_eval_v1.jsonl
- Ability to run Python scripts

# Scenario
# Activity 1: Run the Benchmark Harness

## Objective



Execute the benchmarking harness on the provided dataset (15 prompts) and generate JSONL + CSV outputs.

# Input/Data

datasets/support_chat_eval_v1.jsonl

This dataset contains real user-support prompts.

# Steps to Complete

# Step 1: Inspect the Harness

- Open run_eval.py.
- Locate the sections that:
  - Load the dataset
  - Call the LLM
  - Write JSONL/CSV outputs

# Step 2: Configure the Run

Ensure the script uses:

- The provided dataset
- A single model + region
- Deterministic settings (temperature 0) where possible

# Step 3: Run the Baseline

```
python run_eval.py --dataset datasets/support_chat_eval_v1.jsonl
--variant baseline
```

Confirm outputs appear in the runs/ directory.

# Step 4: Verify Output Files

Check the summary CSV includes:

- prompt ID
- ttft_ms, total_ms
- token counts
- cost per task

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## Expected Outcome:

- A fully completed baseline run
- A CSV file with one row per test case

# Activity 2: Analyze Baseline Metrics in pandas

# Objective

Compute summary stats, outliers,

and build a baseline table.



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## Expected Outcome:

- CSV generated from Activity 1:
  - runs/YYYY-MM-DD_baseline.csv

## Essential imports

```
import pandas as pd
df = pd.read_csv("runs/your_baseline_file.csv")
```

# Steps to Complete

## Step 1: Load Data into pandas

(unchanged)

## Step 2: Inspect Raw Metrics

Confirm columns:

- ttft_ms
- total_ms
- input_tokens, output_tokens
- tokens_per_sec
- cost_per_task

## Step 3: Compute Summary Stats

- p50
- p95
- avg tokens/sec
- avg cost/task

## Step 4: Identify Outliers

- Sort by total_ms descending
- Optionally sort by cost_per_task
- Record 2–3 specific slow requests

## Step 5: Build a Summary Table

- Include:
  - p50/p95 ttft_ms
  - p50/p95 total_ms
  - avg tokens/sec
  - avg cost/task
  - outlier counts

## Expected Outcome:

A clear summary table capturing the health of the baseline.

## Activity 3: Formulate Performance Hypotheses

(unchanged except for dataset references; content was fine)

# Compile Your Findings

1 **Requirement**

**Clear description:**
A reproducible baseline report containing raw metrics for each test case generated from Activity 1.

**Include:**

One CSV file with **one row per prompt**.

Columns for at least:

- prompt_id (or text snippet)
- ttft_ms
- total_ms
- input_tokens
- output_tokens
- tokens_per_sec (or derivable)
- cost_per_task (or derivable)

File should be produced by your run of:

python run_eval.py --dataset datasets/support_chat_eval_v1.jsonl --variant baseline

Ensure outputs are stored inside the runs/ directory with the correct timestamped filename.

2 **Requirement**

**Clear description:**
A structured summary table that aggregates key latency, throughput, and cost metrics across all test cases and highlights notable patterns.

- **Include:**
  - p50 and p95 for:
    - ttft_ms
    - total_ms
  - Average values for:
    - tokens_per_sec
    - cost_per_task
  - Outlier analysis including:
    - 2–3 slowest requests (based on total_ms)

Any unusually high-cost tasks

A brief narrative (2–4 sentences) summarizing what the numbers reveal about overall model behavior.

## 3 Requirement

**Clear description:**
A short narrative analysis connecting your baseline data to performance opportunities.

**Include:**

Two hypotheses about the largest performance wins available.

Each hypothesis must reference **specific metrics** (latency, tokens/sec, cost per task, or outliers).

Explain why each area looks promising for optimization.

The reflection should be approximately 200 words in total.

# Summary

- **Provide a concise 3–5 sentence summary** synthesizing your baseline results:
- What you observed about latency distribution
- How stable throughput and token costs were
- Whether outliers indicate bottlenecks
- Overall readiness of the system for deeper optimization work
- This summary should serve as the executive snapshot of your findings.

# Resources

**Primary Materials:**

- run_eval.py
- datasets/support_chat_eval_v1.jsonl
- datasets/support_chat_eval_v1.csv

**Reference Docs:**

- OpenAI API docs
- pandas docs

# Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

# Instructions for Documenting and Sharing The Project for Peer Review

## 1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

## 2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

# Instructions for Documenting and Sharing The Project for Peer Review

## 3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.